

$(X_1, y_1), \dots, (X_n, y_n) \stackrel{i.i.d}{\sim} X, y$ Σ invertible
 where $X \sim N(0, \Sigma)$ ($\Sigma p \times p$) (p features)
 $y = \langle X, \theta \rangle + \varepsilon$ $E(\varepsilon|X) = 0, E(\varepsilon^2) = \sigma^2$.

X $n \times p$ design matrix, $y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

$$\hat{\beta} = (X^T X)^{-1} X^T y \quad (\text{OLS})$$

$$Z_i = \Sigma^{-1/2} X_i \sim N(0, I_p) \quad (\text{no correlation})$$

Z $n \times p$ matrix, rows: $Z_1, \dots, Z_n$. where

$$X = Z \Sigma^{-1/2}$$

Let (X_{n+1}, y_{n+1}) (test data): $(X_{n+1}, y_{n+1}) \sim (X_1, y_1)$ and

$(X_{n+1}, y_{n+1}) \perp \mathcal{D}_n$.

prediction error:
$$E(y_{n+1} - X_{n+1}^T \hat{\beta})^2 = E(X_{n+1}^T \hat{\beta} - X_{n+1}^T \theta - \varepsilon_{n+1})^2 \quad (*)$$

$$= E(X_{n+1}^T (\hat{\beta} - \theta))^2 + \sigma^2$$

$$= E\left(\underbrace{E(X_{n+1}^T (\hat{\beta} - \theta) | X_{n+1})}_{:= g(X_{n+1})} + \varepsilon_{n+1}\right)^2 + \sigma^2$$

$$:= g(X_{n+1})$$

$$g(x) = E(X^T (\hat{\beta} - \theta) | x) \quad (x \in \mathbb{R}^p)$$

$$= E(E(X^T (\hat{\beta} - \theta))^2 | X_1, \dots, X_n)$$

$$= E\left(0 + \sigma^2 (X^T X)^{-1} x\right) \quad (**)$$

$$\Rightarrow \text{error} = \sigma^2 (E(X_{n+1}^T (X^T X)^{-1} X_{n+1}) + 1)$$

$$\text{Let } Z_{n+1} = \Sigma^{-1/2} X_{n+1} \sim N(0, I_p)$$

$$\text{prediction error} = \sigma^2 (E(Z_{n+1}^T (Z^T Z)^{-1} Z_{n+1}) + 1)$$

so same as when there is no correlation in the features...

details for (*) and (**) on the next page if necessary.

details:

$$\begin{aligned}
 (*) \quad A &:= E \left(X_{n+1}^T (\hat{\theta} - \theta) + \varepsilon_{n+1} \right)^2 \\
 &= E \left(E \left(X_{n+1}^T (\hat{\theta} - \theta) + \varepsilon_{n+1} \right)^2 \mid \mathcal{D}_n \right) \\
 &= E \left(g(\mathcal{D}_n) \right) \quad \text{where } g(\mathcal{D}_n) = E \left(X_{n+1}^T (\tilde{\theta} - \theta) + \varepsilon_{n+1} \right)^2 \quad \text{and } \tilde{\theta} = \hat{\theta}(\mathcal{D}_n) \text{ (fixed)} \quad ((\varepsilon_{n+1}, X_{n+1}) \perp \mathcal{D}_n) \\
 &= E \left(X_{n+1}^T (\tilde{\theta} - \theta) \right)^2 + E \left(\varepsilon_{n+1}^2 \right) + \underbrace{E \left(X_{n+1}^T (\tilde{\theta} - \theta) \varepsilon_{n+1} \right)}_{=0} \\
 &= E \left(X_{n+1}^T (\tilde{\theta} - \theta) \right)^2 + \sigma^2 \\
 \therefore A &= E \left(X_{n+1}^T (\hat{\theta} - \theta) \right)^2 + \sigma^2.
 \end{aligned}$$

$$\begin{aligned}
 (***) \quad B &:= E \left(x^T (\hat{\theta} - \theta) \right)^2 \mid X_1, \dots, X_n \\
 &= \text{'bias'} + \text{var} \quad \text{with deterministic design'} \quad (\text{see NLG}) \\
 &= 0 + \sigma^2 x^T \sum \hat{\theta} x \\
 &= 0 + \sigma^2 x^T (X^T X)^{-1} x.
 \end{aligned}$$